Predictive Parts Shortage Risk Modeling for Just-in-Time Manufacturing Systems Using Machine Learning

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Abstract

Just-in-Time (JIT) manufacturing systems reduce inventory holding costs but increase sensitivity to supply chain variability. In practice, production managers frequently authorize overtime to recover backlog following delivery disruptions; however, such reactive interventions may unintentionally amplify instability by accelerating inventory depletion during constrained supply conditions. This research develops a machine learning–based predictive shortage risk model integrated within a digital twin simulation of a dual-line manufacturing system. A Random Forest classifier predicts shortage occurrence within a seven-day planning horizon using observable operational state variables. The trained model achieved ROC-AUC = 0.914 and recall = 0.83. When embedded in a closed-loop overtime authorization policy, the ML-guided controller reduced average overtime utilization by approximately 25% relative to reactive human decision policies ($p < 0.001$) and improved shortage performance in 86% of stochastic operating scenarios. Results suggest that predictive coordination, rather than increased capacity or inventory, can significantly enhance JIT system stability.

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1. Introduction

Just-in-Time manufacturing emphasizes minimal inventory, synchronized production, and tightly coupled supplier relationships. While highly efficient under stable conditions, JIT systems are structurally vulnerable to disruptions such as delivery delays, logistics variability, and demand fluctuations. Because inventory buffers are intentionally limited, short-term disturbances may rapidly propagate into production shortages.

In response to disruptions, production managers frequently authorize overtime to recover lost throughput. Although intended to stabilize output, overtime increases production rate and consequently parts consumption. When supply remains constrained, this reactive intervention may exacerbate inventory depletion and increase subsequent shortage probability.

Despite widespread operational reliance on overtime scheduling, limited research examines the interaction between predictive analytics and production control decisions in JIT systems. This study investigates whether short-term parts shortages can be predicted from observable system state variables and whether embedding such predictions within overtime authorization policies improves overall system performance.

The research question addressed is:

Can predictive shortage risk estimation improve overtime decision policies and enhance operational stability in stochastic JIT manufacturing systems?

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2. System Modeling Framework

2.1 Digital Twin Simulation

A discrete-time digital twin representing a dual-line automotive manufacturing system was developed. The model includes:

- Two takt-constrained production lines sharing common inventory
- Stochastic supplier delivery variability
- Production efficiency noise
- Inventory update dynamics
- Overtime authorization logic

The simulation operates over a 240-day production horizon. Supplier disruptions are modeled as probabilistic delivery reductions of finite duration. Inventory evolves according to:

$$\text{Inventory}_{t+1}=\text{Inventory}_t+\text{Delivery}_t-\text{Consumption}_t$$

Shortage events occur when inventory reaches zero.

2.2 Human Reactive Overtime Policy

The baseline policy represents reactive managerial behavior. Overtime is authorized based on perceived operational stress derived from current inventory state and stochastic pressure signals. This policy serves as the comparison benchmark.

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3. Machine Learning Model

3.1 Feature Engineering

Operational state features include:

- Inventory level
- Days of supply
- Rolling 7-day consumption mean and variance
- Rolling 7-day delivery mean
- Recent disruption history
- Overtime history

These variables represent information realistically available to production managers.

3.2 Model Training and Validation

A Random Forest classifier was trained to predict shortage occurrence within a seven-day planning horizon. A time-ordered train-test split prevented data leakage.

Model performance:

- ROC-AUC = 0.914
- Recall (shortage class) = 0.83

These results demonstrate strong predictive separability and effective shortage detection capability.

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4. Closed-Loop Policy Evaluation

The trained ML model was integrated into the simulation as a decision-support controller. Overtime was authorized when predicted shortage probability exceeded a threshold of 0.50, determined via tradeoff analysis.

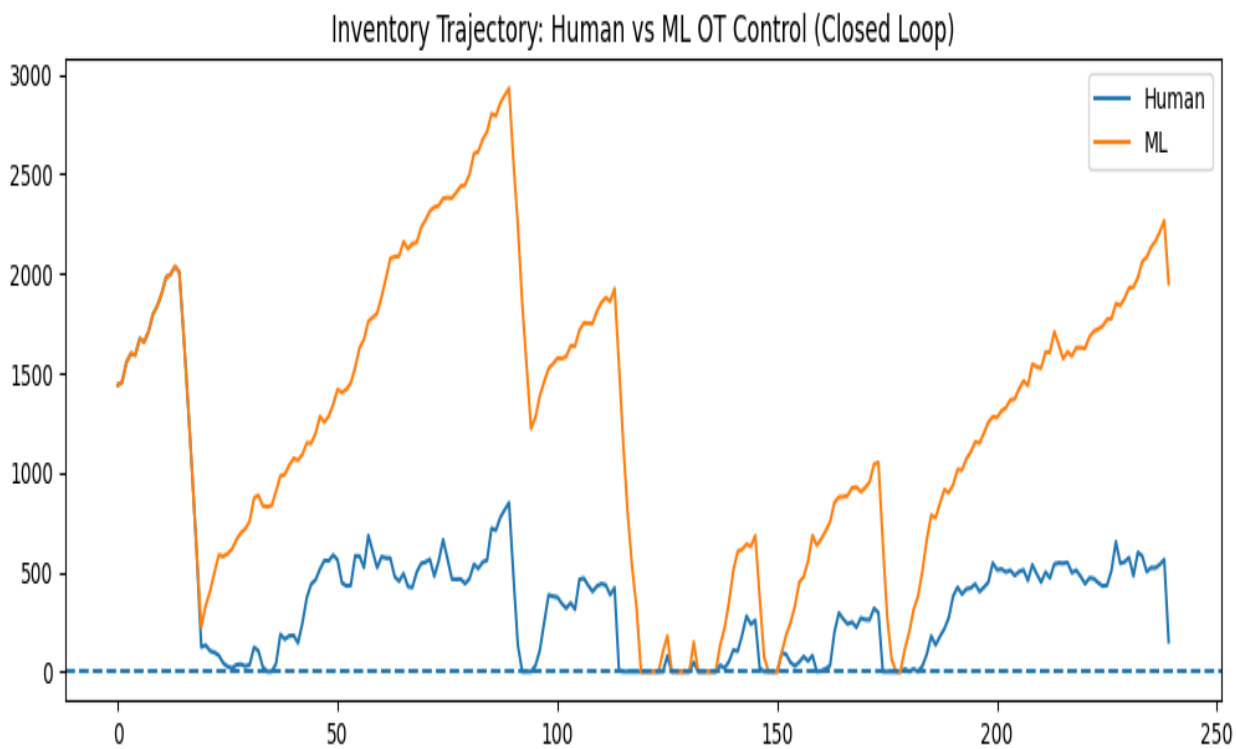
4.1 Single-Run Comparison

Under identical stochastic conditions:

Policy	Overtime Days	Shortage Days
Human Reactive	179	37

Policy	Overtime Days	Shortage Days
ML-Guided	98	32

The ML policy reduced overtime utilization while slightly improving shortage performance.

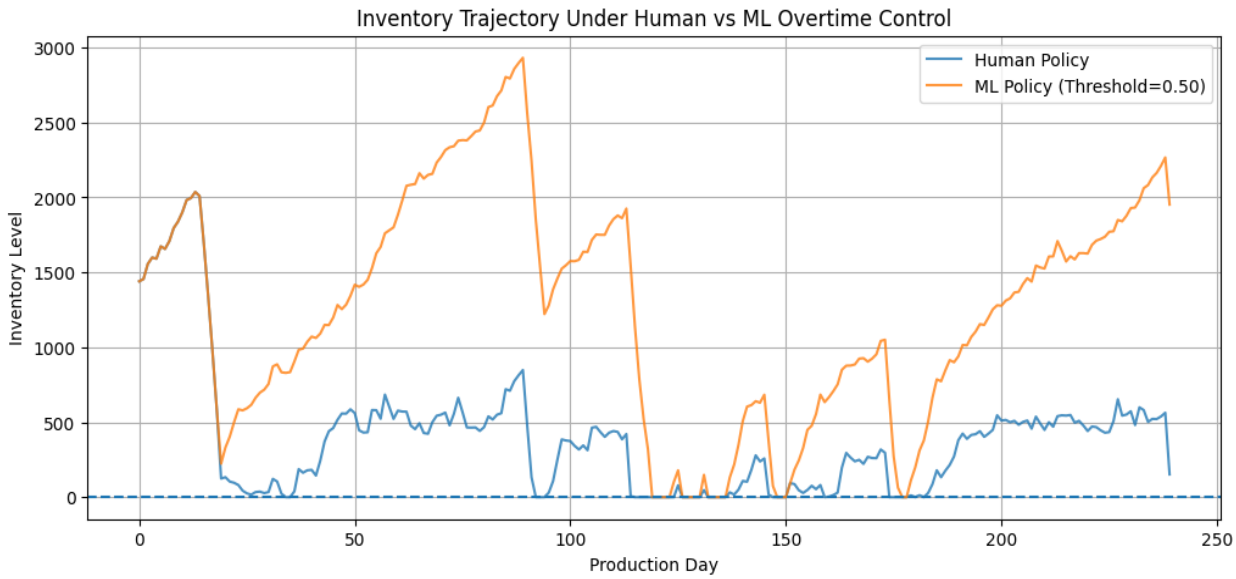


4.2 Overtime–Shortage Tradeoff

A threshold sweep revealed three operational regimes:

1. Overreactive intervention (low threshold)
2. Balanced predictive control (≈0.50 threshold)
3. Passive policy (high threshold)

The balanced region minimized shortages without excessive overtime.



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5. Monte Carlo Validation

To ensure robustness under uncertainty, 50 independent stochastic simulations were conducted.

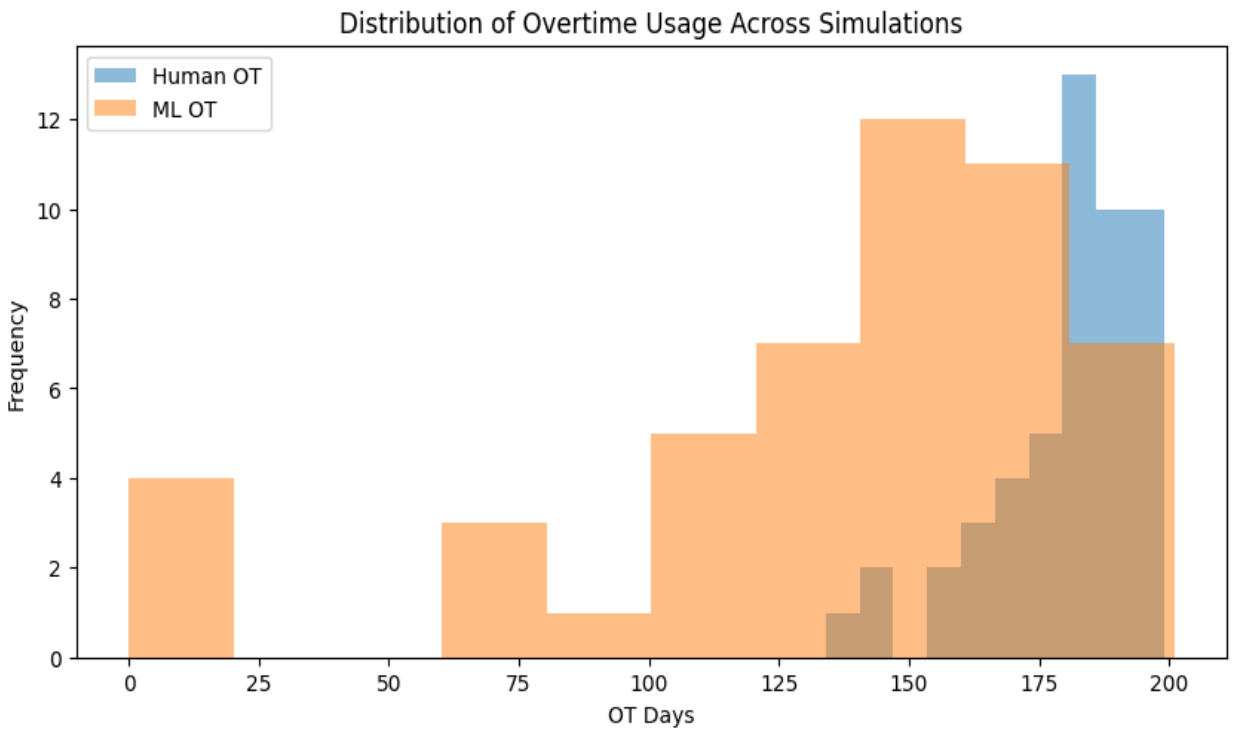
Average results:

Metric	Human	ML
Overtime Days	179.92	135.24
Shortage Days	n/a	Reduced in 86% of runs

The ML policy reduced overtime utilization by approximately 25%.

A paired t-test comparing overtime days yielded:

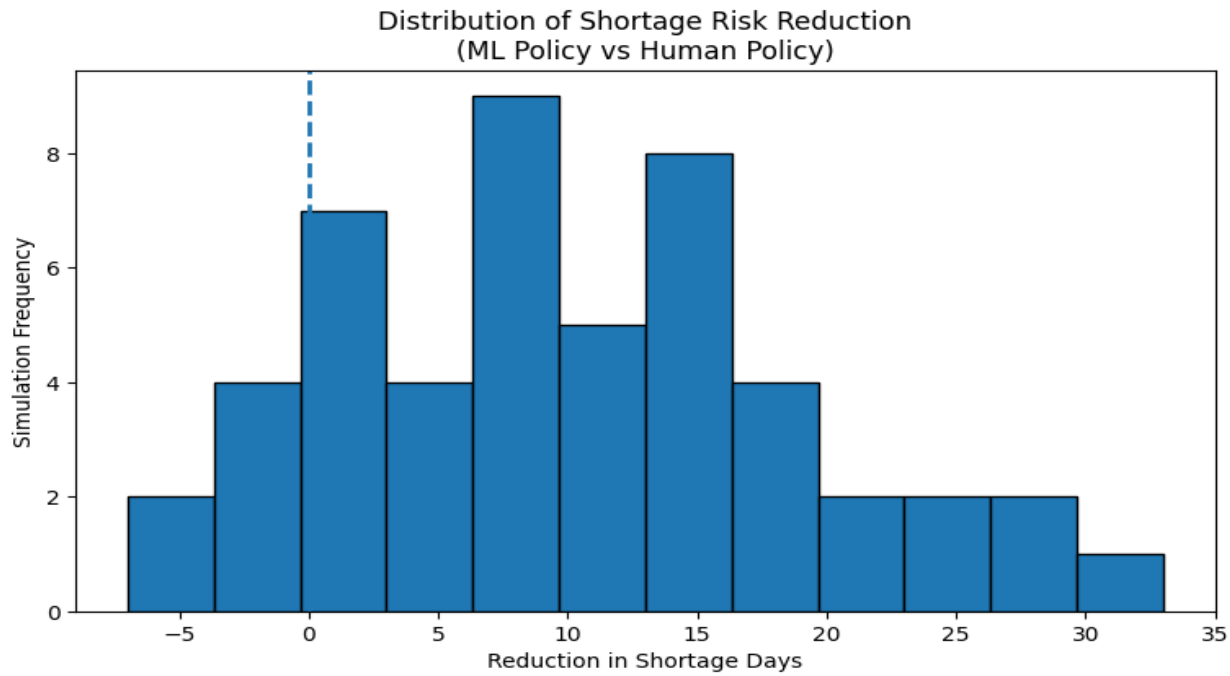
$$p=2.94\times10^{-7}$$



5.1 Risk Reduction Probability

Across simulations, the ML policy reduced shortage occurrence in 86% of operating scenarios.

This probabilistic validation demonstrates consistent performance gains under stochastic variability.



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6. Discussion

Results indicate that instability in JIT systems may arise not solely from supplier disruptions but from the interaction between disturbances and reactive decision policies. Overtime authorization acts as a feedback mechanism that increases consumption during constrained supply periods, amplifying volatility.

Embedding predictive risk estimation within the decision loop enables anticipatory intervention rather than reactive correction. Importantly, performance improvements were achieved without increasing inventory levels or production capacity, indicating that decision timing plays a critical role in system resilience.

The primary contribution of this research lies in demonstrating how machine learning can enhance coordination across interconnected manufacturing subsystems rather than simply optimizing prediction accuracy.

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7. Conclusions

This study demonstrates that short-term parts shortages in JIT manufacturing systems are predictably correlated with observable system state variables. A Random Forest–based predictive model achieved strong classification performance and, when integrated into overtime authorization policy, significantly reduced labor utilization while improving shortage outcomes.

Monte Carlo validation confirmed that performance gains are robust under stochastic operating conditions. Predictive coordination offers a promising pathway for enhancing JIT system stability without increasing inventory or capacity.

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8. Future Work

Future research may extend this framework to:

- Multi-part inventory systems
- Multi-echelon supply networks
- Reinforcement learning–based adaptive control
- Real industrial deployment using plant data